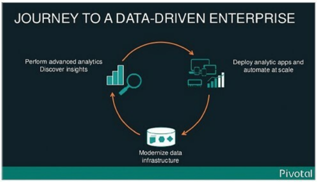

Figure 1: Journey for a data driven enterprise (Image credits: Google Images)

2. **Variety:** The type and nature of the Big Data helps people who analyse it to effectively use the resulting insights.
3. **Velocity:** Big Data is also identified by the rate at which the data is generated and processed to meet various demands.
4. **Variability:** We can consider a data set to be Big Data if it's not consistent, hampering various processes that are used to handle and manage it.
5. **Veracity:** In some sets of data, the quality varies greatly and it becomes a challenging task to analyse such sets, as this leads to a lot of confusion during analysis.

The various challenges associated with such large amounts of data include:

1. Searching, sharing and transferring
2. Curating the data
3. Analysis and capture
4. Storage, updation and querying
5. Information privacy

How enterprises began leveraging Big Data

Considering the tremendous increase in the demand for various online enterprise applications nowadays, the present era could well be named the *Enterprise Era*. This is best illustrated by the fact that around 1 million transactions per hour are being tracked by Walmart. This statistic makes one ponder over how difficult it has become for various enterprise applications to track and use such mammoth amounts of unstructured data.

Clearly, effectively using data can be a difficult task, specially with the increasing number of new data sources, the requirements for fresh data, and the need for increased processing speed. Hence, for advanced operational efficiencies and accelerated business growth, enterprises need to address and overcome these challenges. Various Big Data techniques and methodologies are being adopted to process and get the Right Data (that which is sufficient and appropriate for use) out of such unstructured data sets.

In the recent past, many enterprises have invested heavily on developing various data warehouses. These can serve as the central data system to report, extract, transform and load different processes, and also ingest data

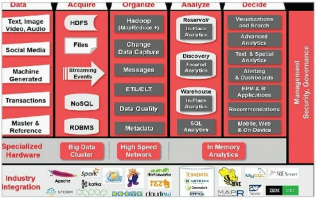

Figure 2: Architecture implemented by enterprises for Big Data (Image credits: Oracle Big Data Guide)

from different databases and other sources—both inside and outside the enterprise. Since the variety, velocity and volume of data continues to increase, it is overloading such expensive enterprise data warehouses, causing a significant processing burden.

To get rid of this bottleneck, organisations are opting for different open source tools like Hadoop to offload data warehouse processing functions. Hadoop can help organisations lower costs and turn highly efficient if it's being used along with various data warehouses. However, as Hadoop requires some special skillsets to deploy it, organisations have started trying out other alternatives. A solution developed by the combined efforts of Dell, Intel, Cloudera and Syncsort works on the use-case driven Hadoop Reference Architecture. This technology simplifies data processing with the help of an architecture, which helps users to optimise an already existing data warehouse. This offloading solution provides a Hadoop environment using Cloudera Enterprise software. The Cloudera Distribution of Hadoop (CDH) delivers all the core elements of Hadoop, like scalable storage and distributed computing. It allows users to reduce the Hadoop deployment period to just several weeks, develop Hadoop jobs within hours, and become completely productive. CDH also ensures high availability, security and integration with the large set of other tools.

The Big Data enterprise model

Let's have an overview of the general Big Data model that enterprises are implementing, which mainly consist of several intermediate systems or processes that are featured below.

Data source: These are the datasets on which different Big Data techniques are implemented. They can exist in an unstructured, semi-structured or structured format. There are unstructured datasets which are extracted from several social media applications in the form of images, audio/video clips or text. The semi-structured datasets are generated by different machines and require less effort to convert them to the structured form. Some data sets are already in the structured